In each of the panels in Figure 2.14:

- The solid line in the middle is the mean function, which is analogous to the solid line connecting the diamonds in Figure 2.11.
- The shaded region, on the other hand, is the 95% CI across the domain, corresponding to the error bars in Figure 2.11.

Before observing any data, we start out with the *prior GP* in the top left panel. Just like a prior multivariate Gaussian, our prior GP produces constant mean prediction and uncertainty in the absence of training data. This is a reasonable behavior to have.

The interesting part comes when we condition our GP on various data points. This is visualized in the remaining panels of Figure 2.14. Exactly like what we see in the discrete case of a multivariate Gaussian, with a GP working in a continuous domain, the mean prediction as well as samples drawn from the posterior distribution smoothly interpolate data points in the training set, while our uncertainty about the function value, quantified by the CI, smoothly decreases in the areas around these observations. This is what we call a *calibrated quantification of uncertainty*, which is one of the biggest selling points of a GP.

SIDEBAR **Smoothness of a Gaussian process**

The smoothness property refers to the constraint that requires similar points to be correlated with each other. In other words, points that are similar should result in similar function values. This is, again, why when we condition on the data point at 3 in the top right panel, the mean predictions at 2.9 and 3.1 are updated to take on larger values than their prior means. This smoothness is set using the covariance function of the GP, which is the topic of Chapter 3.

While the examples we have seen so far is in one dimension, this smoothness is preserved when our search space is higher-dimensional, as we will see later.

Overall, we have seen that a GP is a multivariate Gaussian distribution when extended to infinite dimensions, and thanks to many convenient mathematical properties of Gaussian distributions, a GP not only produces a mean prediction but also quantifies our uncertainty about the function values in a principled way via its predictive covariances. The mean prediction exactly go through the training data points, and the uncertainty collapses at these data points.

2.4 Implementing Gaussian processes in Python

In this last section of the chapter, we will take our first step in implementing GPs in Python. Our goal is to become familiar with the syntax and API of the libraries we will be using for this task, and learn how to recreate the visualizations we have seen thus far. This hands-on section will also help us understand GPs more deeply.

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